

EEE522 Autonomous Vehicles

Automatic Emergency Braking

Ivan Ho

Acknowledgements

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This subject is a collaborative development with significant contributions from:

Hongrui Zheng (lead), Matthew O'Kelly (lead), Johannes Betz (lead), Houssam Abbas, Joseph Auckley, Madhur Behl, Luca Carlone, Jack Harkins, Paril Jain, Kuk Jang, Paritosh Kelkar, Sertac Karaman, Dhruv Karthik, Nischal KN, Thejas Kesari, Matthew Lebermann, Kim Luong, Yash Pant, Varundev Shukla, Nitesh Singh, Siddharth Singh, Nandan Tumu, Zirui Zang, and many others.

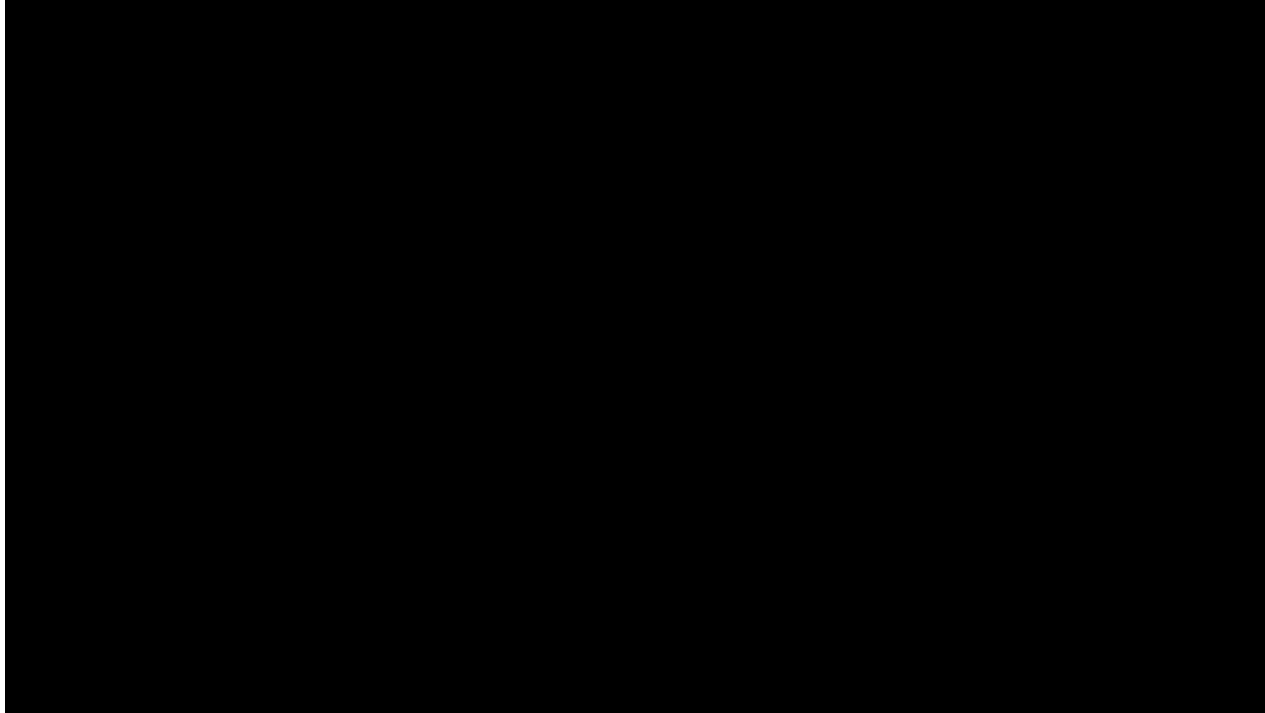
Outline

- Automatic Emergency Braking (AEB) on the road
- Range sensors for autonomous vehicles
- Working with laser scan data
- Measuring safety

AEB on the road

Automatic Emergency Braking (AEB)

Stop the vehicle before you collide with an obstacle...



Source: Nissan USA 5

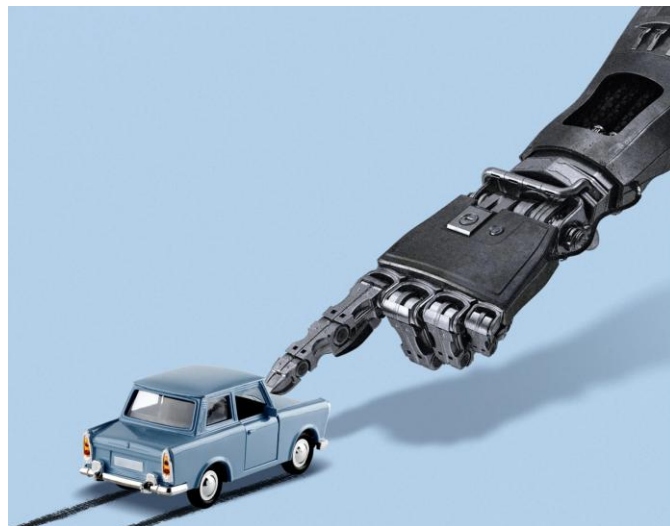
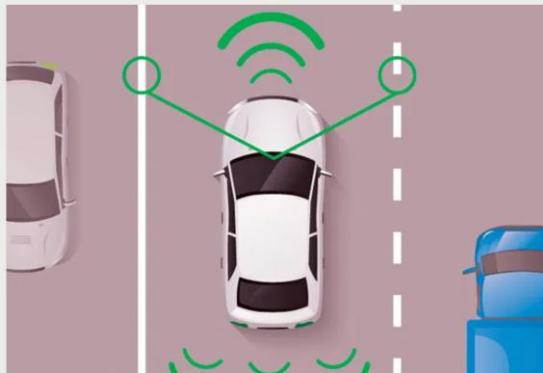
EUROPE MANDATES AUTOMATIC EMERGENCY BRAKING

Safety advocates project it could save 4,000 lives per year

NEWS TRANSPORTATION

Vehicle Safety Ratings Will Soon Include Marks for Crash Avoidance Tech › Starting in 2018, U.S. vehicle safety ratings will account for technologies meant to keep cars from crashing

BY JEREMY HODG | 11 DEC 2018 | 3 MIN READ |

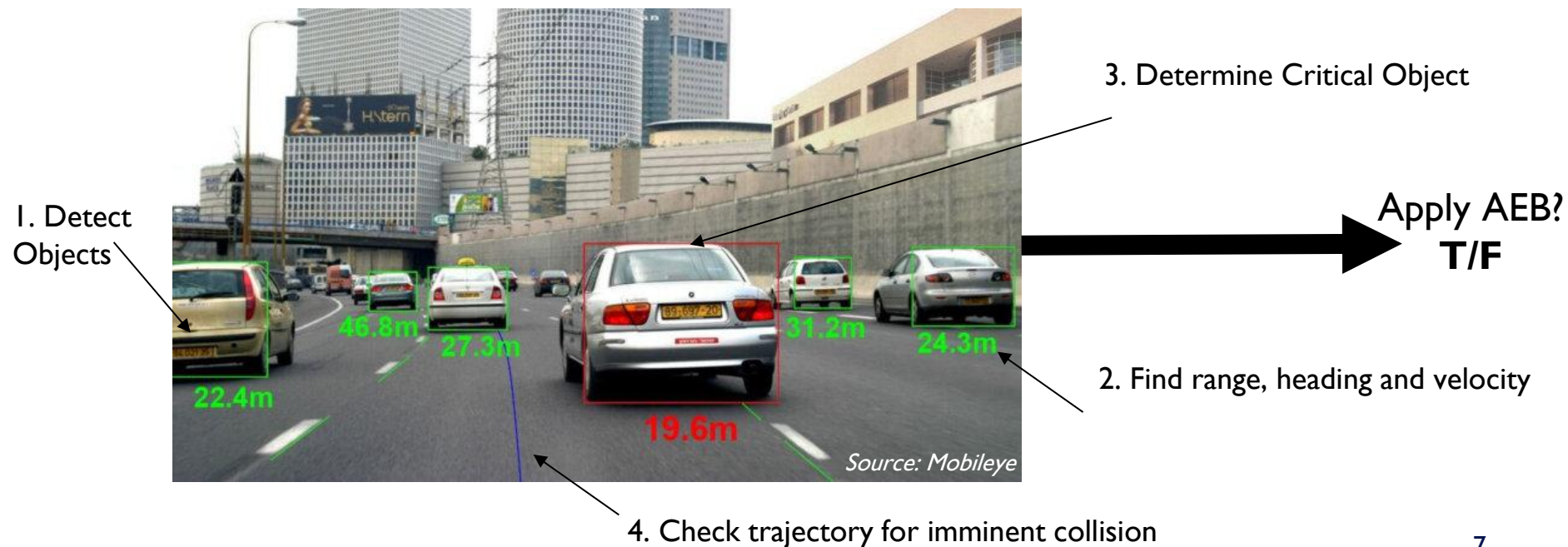


crash-avoidance technologies

- forward collision warning
- crash imminent braking
- dynamic brake support
- lane departure warning
- rollover resistance
- blind spot detection.

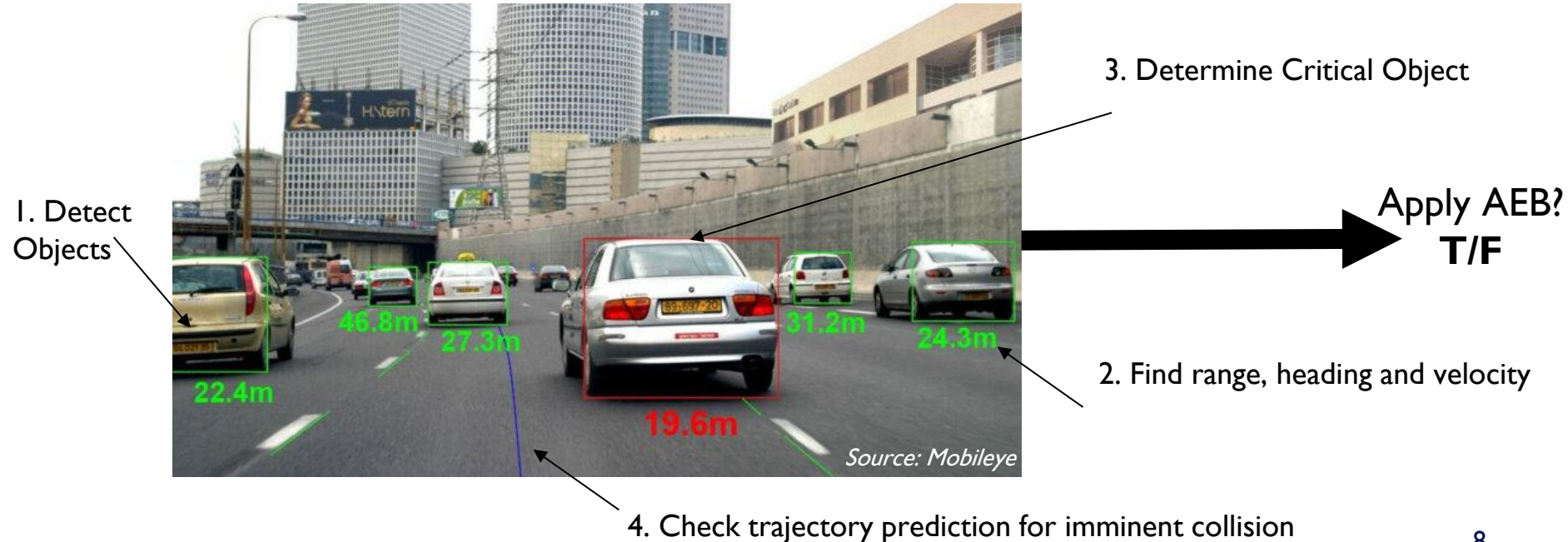
AEB in Today's Vehicles

Fuse radar and calibrated vision pipeline (object detection, tracking, visual odometry)...



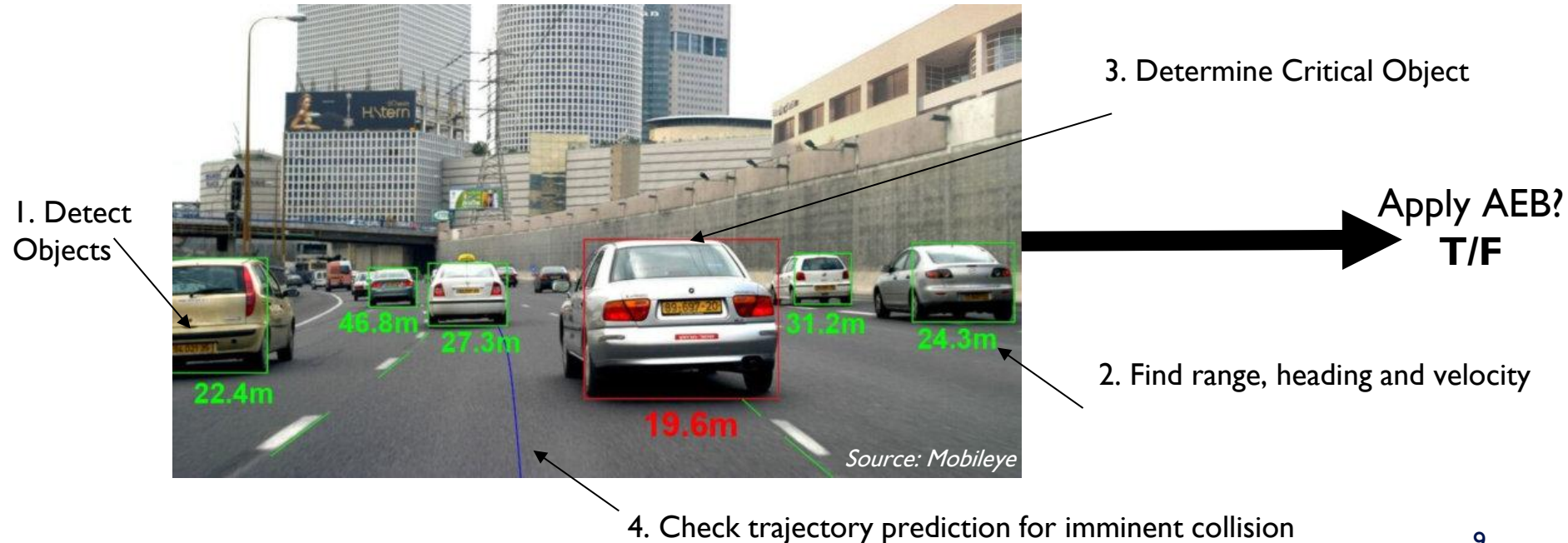
Quiz: AEB in Today's Vehicles

What type of model is this system? Regression? Deep Network? Or something simpler?



Answer: AEB in Today's Vehicles

What type of model is this? **Binary Classifier.**



Quiz: Types of Failures

Can you fill in this table?

		Ground Truth	
		True	False
Classifier Output	True		
	False		

Quiz: Types of Failures

Which one of these is a serious problem?

		Ground Truth	
Classifier Output	True	True Positive	False Positive
	False	False Negative	True Negative

Answers: Types of Failures

Quiz: Which one of these is a serious problem?

It's not this...

Classifier Output	Ground Truth	
	True	False
True	True Positive	False Positive
False	False Negative	True Negative

Engineering incentives aligned.

But... no one will buy a system with false positives.

Answers: Types of Failures



Answers: Types of Failures

Quiz: Which one of these is a serious problem?

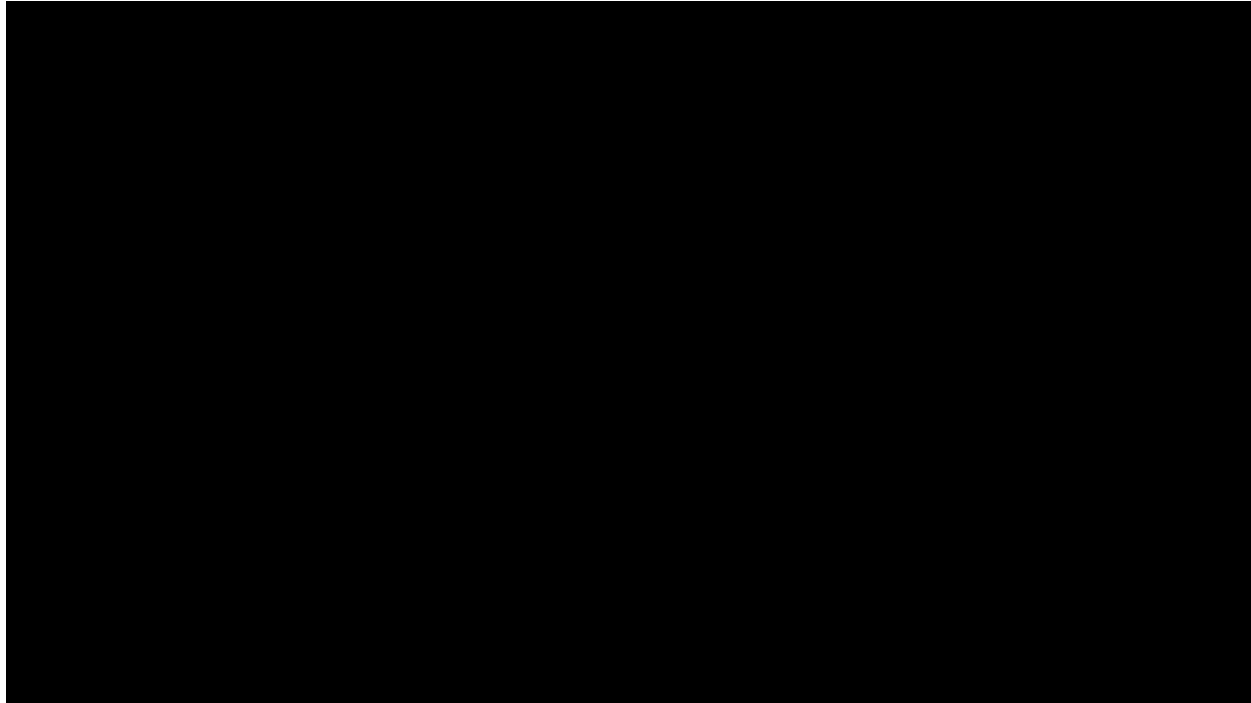
Ground Truth	
Classifier Output	
	True Positive
	False Positive
	False Negative
	True Negative

False negatives kill innocent people. Regulatory & insurance provide incentives.

Types of Failures: False Negative

Uber autonomous car kills a pedestrian

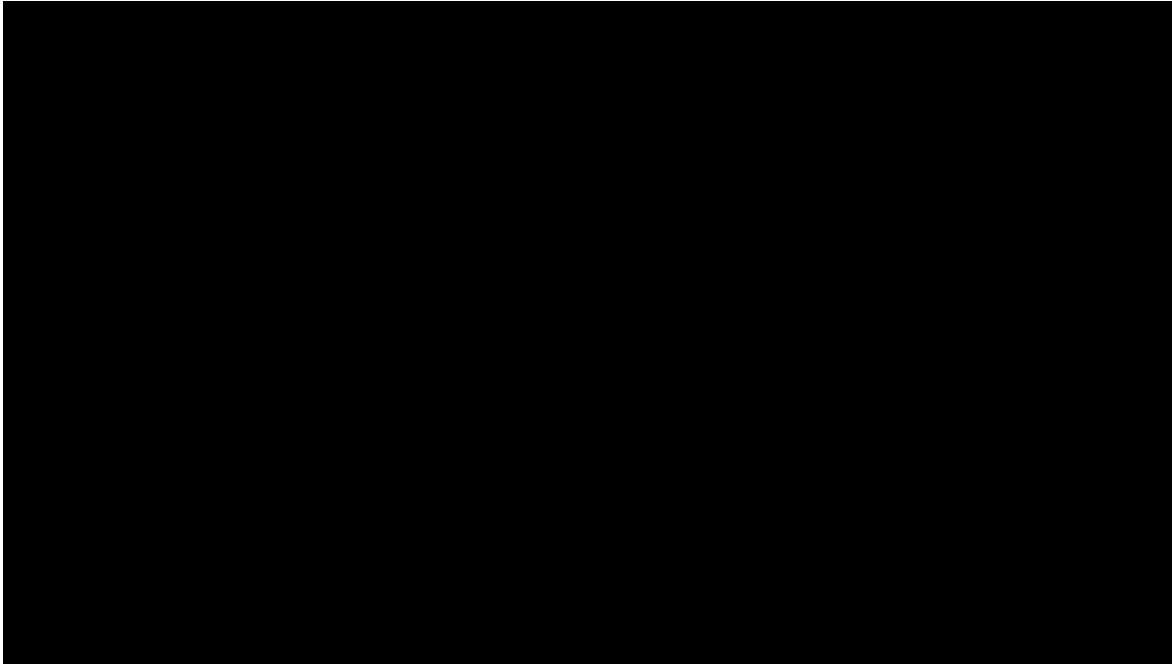
Viewer discretion is advised



Source: The Guardian 15

Is this problem solved?

AEB could have stopped the vehicle, but was turned off because the vehicle had frequent false positives.



Source: Unknown

False Positives & False Negatives

You will need to tune this...

Ground Truth	
Classifier Output	HIGH
	LOW
HIGH	True Positive
LOW	False Positive
Classifier Output	HIGH
	LOW
HIGH	False Negative
LOW	True Negative

Addressing AEB false negatives in the Automotive Industry

Vehicle System Performance Criteria	◆	◆◆	◆◆◆	◆◆◆◆	◆◆◆◆◆
Max Speed	10 mph	30 mph	50 mph	65 mph	75 mph
Detection Reliability	Frequent false negatives or positives	Fewer false negatives or positives	Occasional false negatives or positives	Extremely rare false positives or negatives	Extremely rare false positives or negatives
Pedestrian, Bicycle, Motorcycle Detection and Classification Range	Near range detection, no classification	Mid-range detection	Mid-range detection and classification	Mid-range detection and classification	Far-range detection and classification
Vehicle Detection and Classification Range	Near range detection, no classification	Mid-range detection	Mid-range detection and classification	Far-range detection and classification	Far-range detection and classification
Stationary Object Detection and Classification Range	May not detect	May not detect	Near range detection	Mid-range detection and classification	Far-range detection and classification
Peripheral Object Detection and Classification Coverage	No perception coverage	Limited perception coverage	Limited perception coverage	Improved perception coverage	Excellent perception coverage
Dynamic Driving Intervention Capability	AEB for crash mitigation	Forward collision avoidance AEB	Forward collision avoidance with steer and brake control	Forward and side collision avoidance with steer and brake control	Forward and side collision avoidance with steer and brake control
Light and Weather Performance	Suffers in low light and inclement weather	Better in low light and inclement weather	Better in low light and inclement weather	Performs well in all light and most weather	Performs well in all light and most weather

ADAS Rating		
2.6	◆◆◆◆◆	• Reduced drift with fewer driver interventions (2.5) • Requires lane lines (2.0) • Mid-range lane line detection (2.0) • Better active assistance, some response to peripheral activity (2.0) • Requires flat and straight roads (2.5) • Better in low light and inclement weather (2.0)
Lane Keep Assistance Assists with steering to maintain vehicle within driving lane.*		
2.4	◆◆◆◆◆	• Max speed 40 mph (2.5) • Fewer false negatives or positives (2.0) • Mid-range pedestrian/bicycle/motorcycle detection (2.4) • Mid-range vehicle detection (2.4) • Near-range stationary object detection (3.0) • Limited peripheral perception coverage (3.1) • Forward collision avoidance AEB (2.0) • Suffers in low light and inclement weather (1.0)
Automatic Emergency Braking Detects potential collisions while traveling forward, provides forward collision warning and automatically applies the brakes to avoid or lessen the severity of impact. Includes pedestrian or other object detection.*		
3.2	◆◆◆◆◆	• Good deceleration response (3.7) • Excellent acceleration response (4.1) • Freeway only (2.0) • May not detect stationary objects (3.2) • Limited peripheral perception coverage (3.3) • Better in low light and inclement weather (2.0)
Adaptive Cruise Control Assists with acceleration and/or braking to maintain a prescribed distance between it and a vehicle in front. Can come to a stop and continue.*		
2.9	◆◆◆◆◆	• Wider peripheral field of view (3.0) • Mid-range peripheral object detection (3.5) • Mid-range peripheral vehicle detection (3.4) • Warning only (2.0) • Better in low light and inclement weather (2.0)
Blind Spot Warning Detects vehicles to rear in adjacent lanes while driving and alerts the driver to their presence.*		

Example car sales labels

Range sensors for AVs

Range Sensor Overview



Depth Camera



Stereo Camera



Monocular Camera



Radar



Ultrasonic



Planar LIDAR

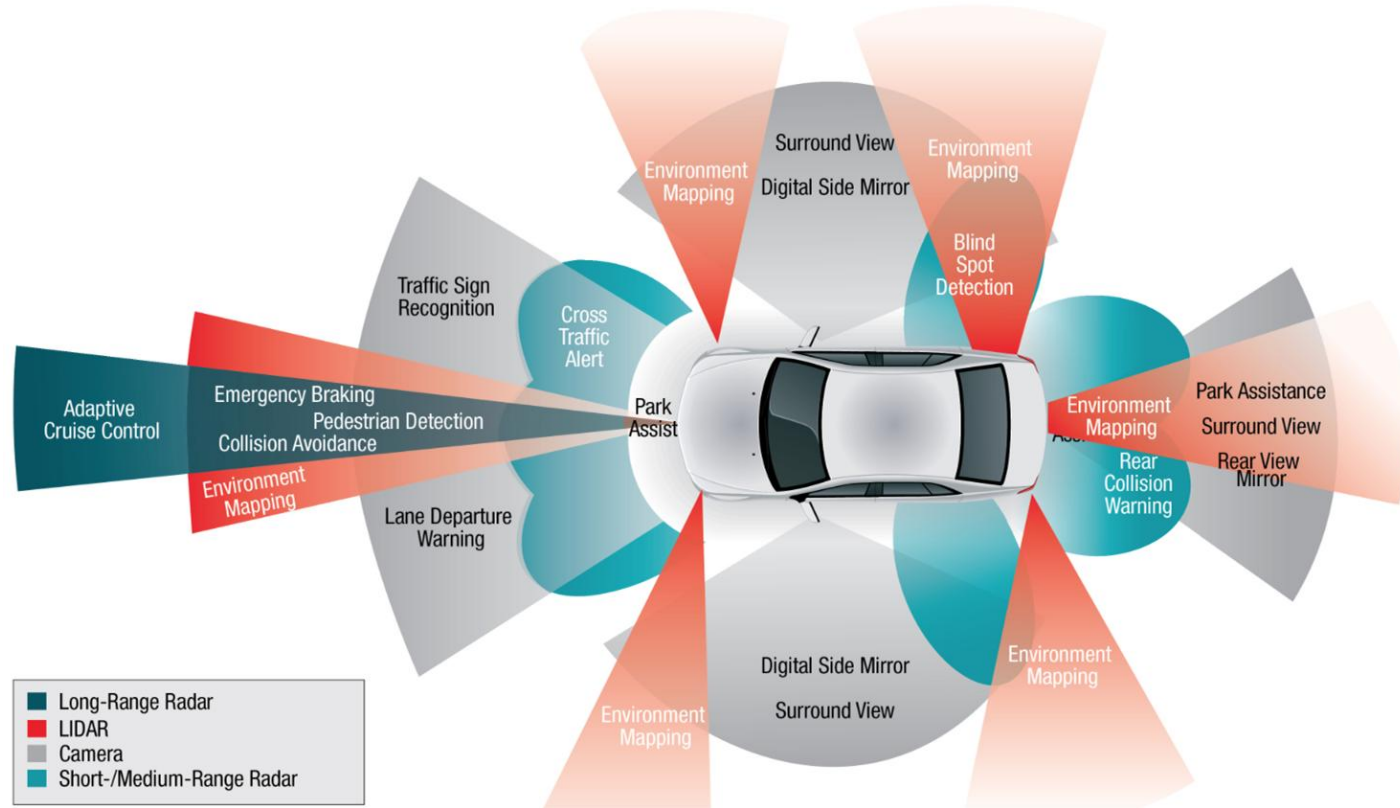


3D LIDAR



Solid State LIDAR

Range Sensor Suite on a Typical AV



Source: Vivota.com

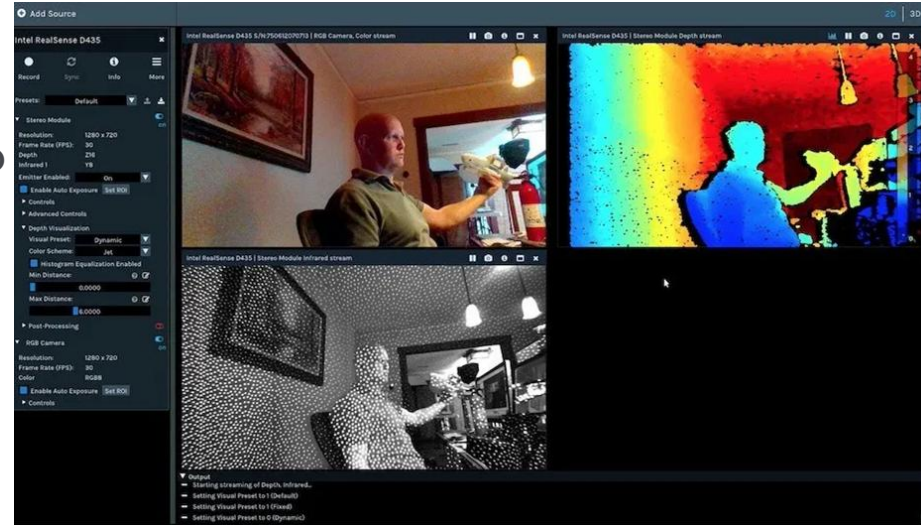
RGB-D Camera

Demo Hardware: Intel Realsense D435i

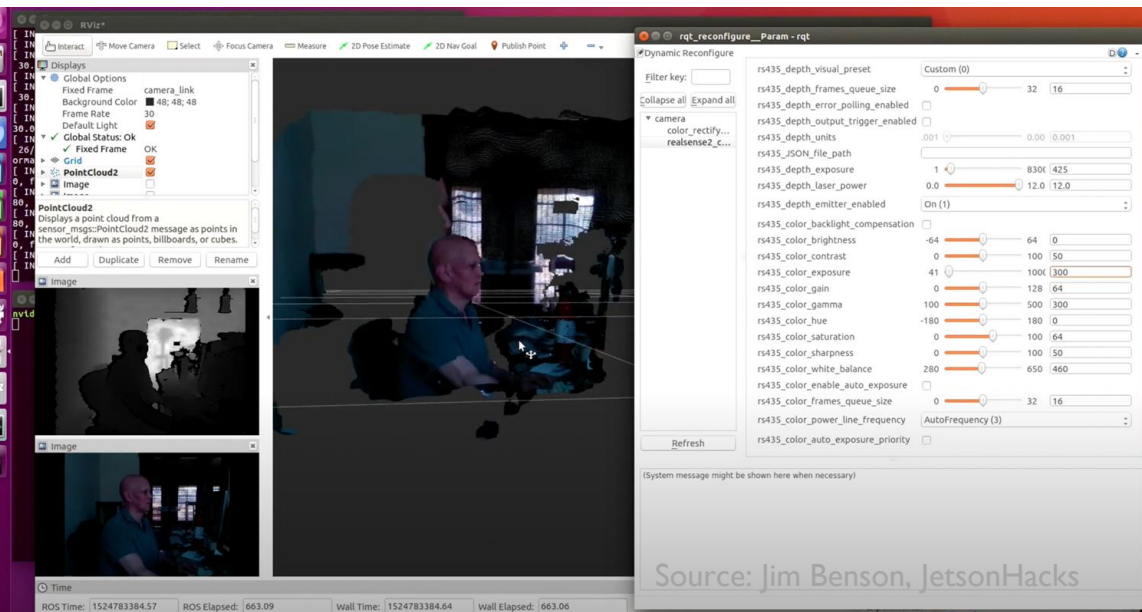
Working Principle: Active Infrared Stereo

Advantages: 3D point cloud - great for visual SLAM, IR Stereo works in any environment, Mature SDK, widely used, calibrated IMU

Disadvantages: Limited range, can be noisy, lighting conditions affect performance, requires building Linux kernel



Realsense RGB-D Camera



About how to use Realsense D435i on F110 cars

Team SAZ

November 29, 2019

1 Keypoints

To use the Realsense D435i camera on F110 cars, we need to:

1. Recompile the Linux kernel
2. Install Realsense 2 SDK
3. Install ROS wrapper for Realsense

2 Details

1. Intel doesn't support realsense on Jetsons so we need patch some video driver in Linux kernel to let the camera function correctly. A guy at JetsonHacks wrote scripts for doing this and it's been tested to work well.

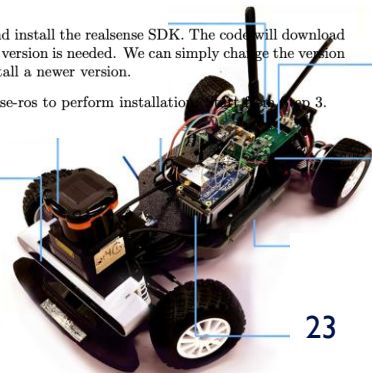
Follow the scripts from <https://github.com/jetsonhacks/buildLibrealsense2TX>.

Run `./buildPatchedKernel.sh` to recompile the kernel.

This script is written for L4T version up to 28.2.1. Make sure the battery is charged and the jetson clocks .sh is used.

2. We can use `./installLibrealsense.sh` to download and install the realsense SDK. The code will download SDK version 2.13.0 but for some cameras, a higher version is needed. We can simply change the version number in `./installLibrealsense.sh` to 2.30.0 to install a newer version.
3. Follow <https://github.com/IntelRealSense/realsense-ros> to perform installation.

RGB Depth
Camera



Stereo Camera



ZED 2 Stereo Camera

Producer: Stereolabs

Type: RGB Stereo

Price: \$449

Range: 0.2-20m

Depth Resolutions:

- 672*376@100 FPS
- 1280*720@60 FPS
- 1920*1080@30 FPS
- 2208*1242@15 FPS

Color Resolutions:

- 672*376@100 FPS
- 1280*720@60 FPS
- 1920*1080@30 FPS
- 2208*1242@15 FPS

Observation:

- Needs feature rich environment
- SDK contains some MV services
- Long range

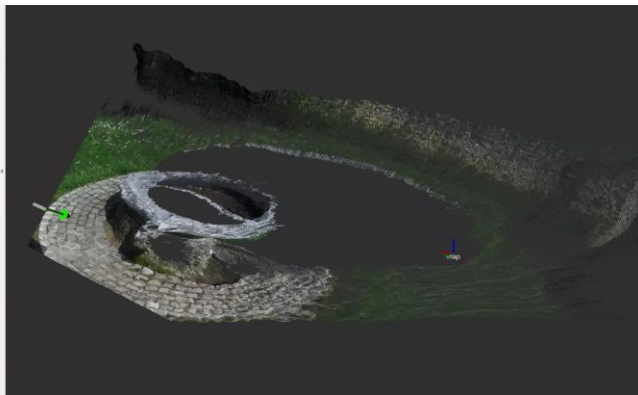
Size: 175*30*33 mm

Demo Hardware: ZED Camera

Working Principle: Stereo/Multiview Geometry

Advantages: Range, outdoor performance, 'scalability', calibrated IMU

Disadvantages: Low texture surfaces, baseline width determines range, processing requirements.



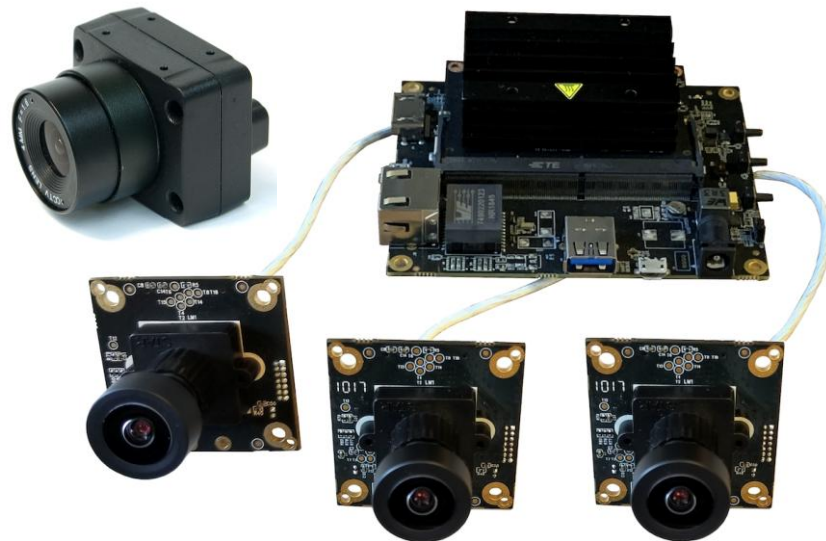
Monocular Depth

Example Hardware: Logitech Webcam & Monodepth Net

Working Principles: Learn from stereo camera data (unsupervised) to map monocular to depth.

Advantages: No special hardware, orthogonal failure modes to other options

Disadvantages: Relative poor accuracy (good for near vs. far questions).



RADAR

Example Hardware: Continental (etc)

Working Principles: Emit RF energy and measure 'echo', can use Doppler shift etc. to get velocity.

Advantages: Cheap, long range, orthogonal failure modes.

Disadvantages: Poor spatial resolution and field of view, false positives on overhead signs etc.



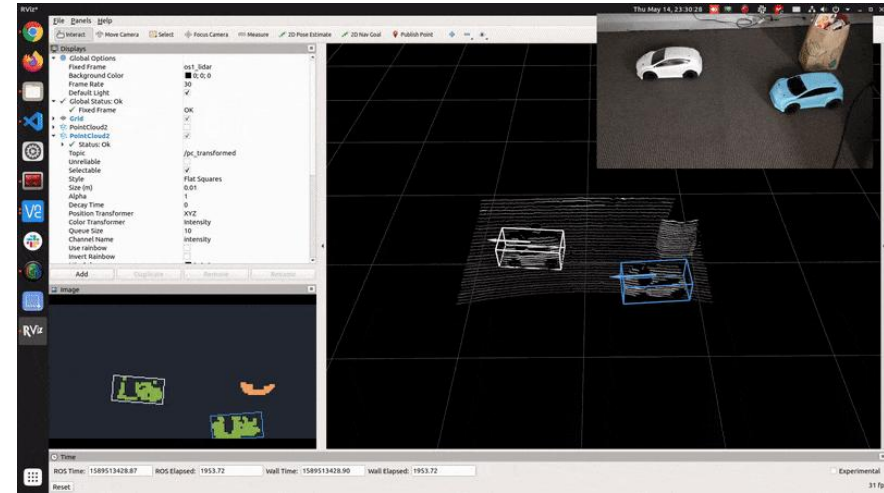
Ultrasonic Proximity Sensor

Example Hardware: VEX IQ Ultrasonic Distance Sensor

Working Principle: Measures time of flight of high-frequency sound waves.

Advantages: Accuracy, cost, size

Disadvantages: Range, resolution, only good enough for parking applications (e.g., 5 m)



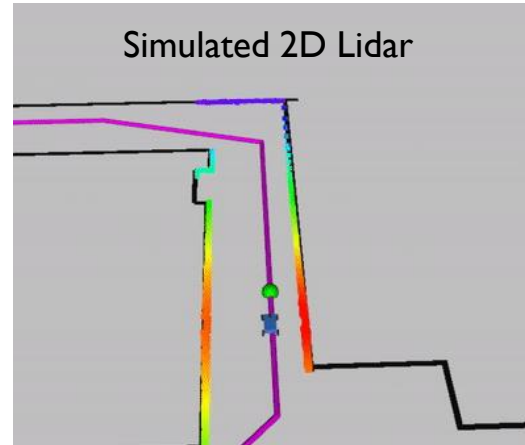
Planar 2D LIDAR

Example Hardware: Hokuyo 30LX

Working Principle: Time-of-flight using laser.

Advantages: Relatively low cost, simple data structure, high update rate, low processing requirements

Disadvantages: Primarily working in flat environments, harder to detect objects etc.



3D LIDAR

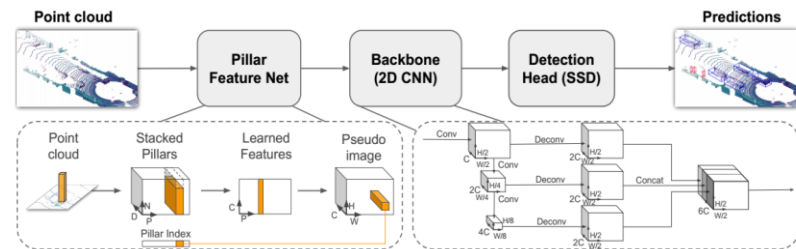
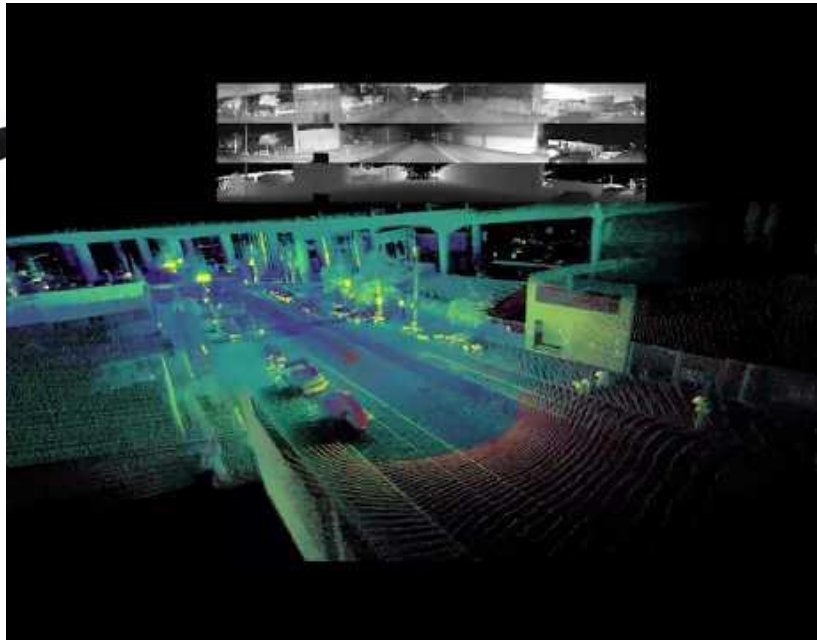
Demo Hardware: Ouster OS-2



Working Principle: Time-of-flight, laser, note different wavelength of this product

Advantages: Full 3D information, 360-degree, can get image like information

Disadvantages: Cost, reliability (mechanical), processing point cloud.



PointPillars: Fast encoders for object detection from point clouds

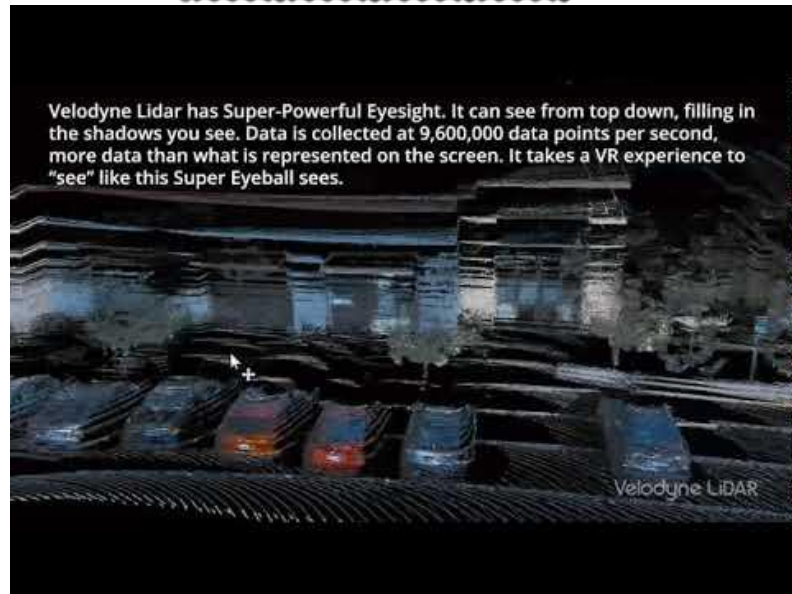
Solid-State LIDAR

Demo Hardware: Velodyne Velarray

Working Principle: Time-of-flight, steer laser with solid-state components.

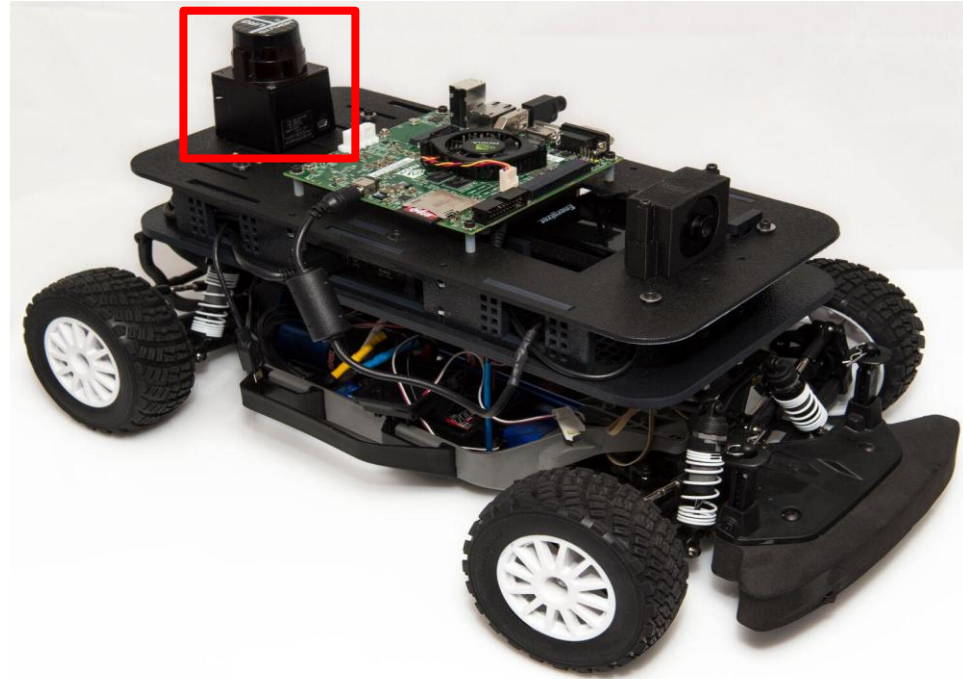
Advantages: Compact, no moving parts, range, etc.

Disadvantages: Field of view, availability, technical feasibility.



Working with Laser Scan Data

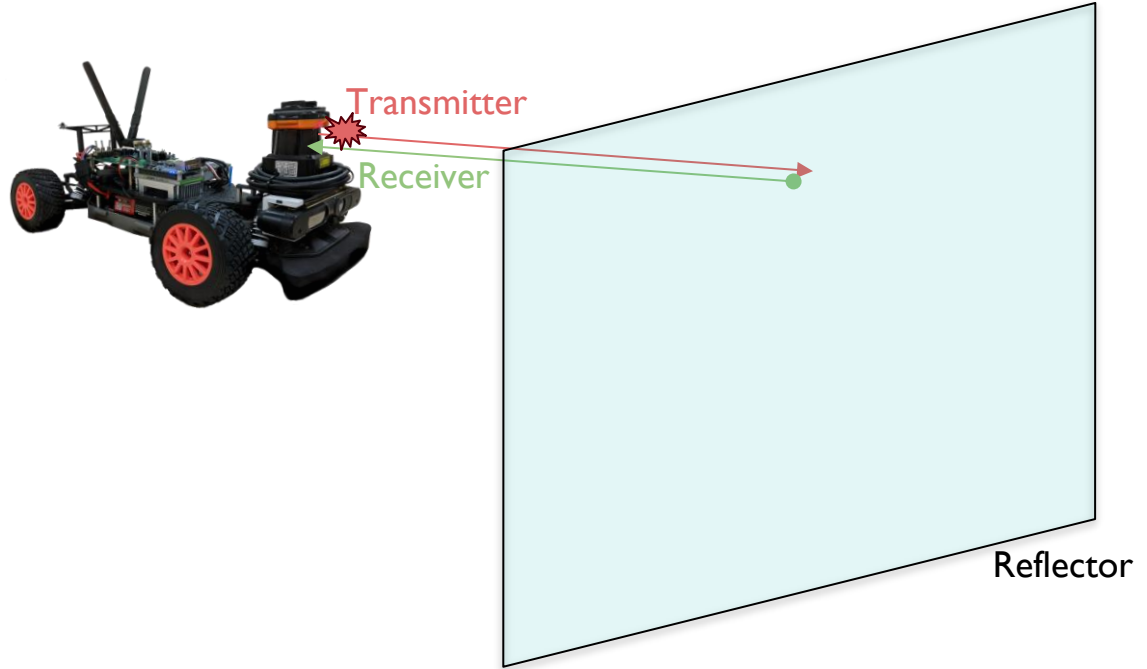
LIDAR on FITenth



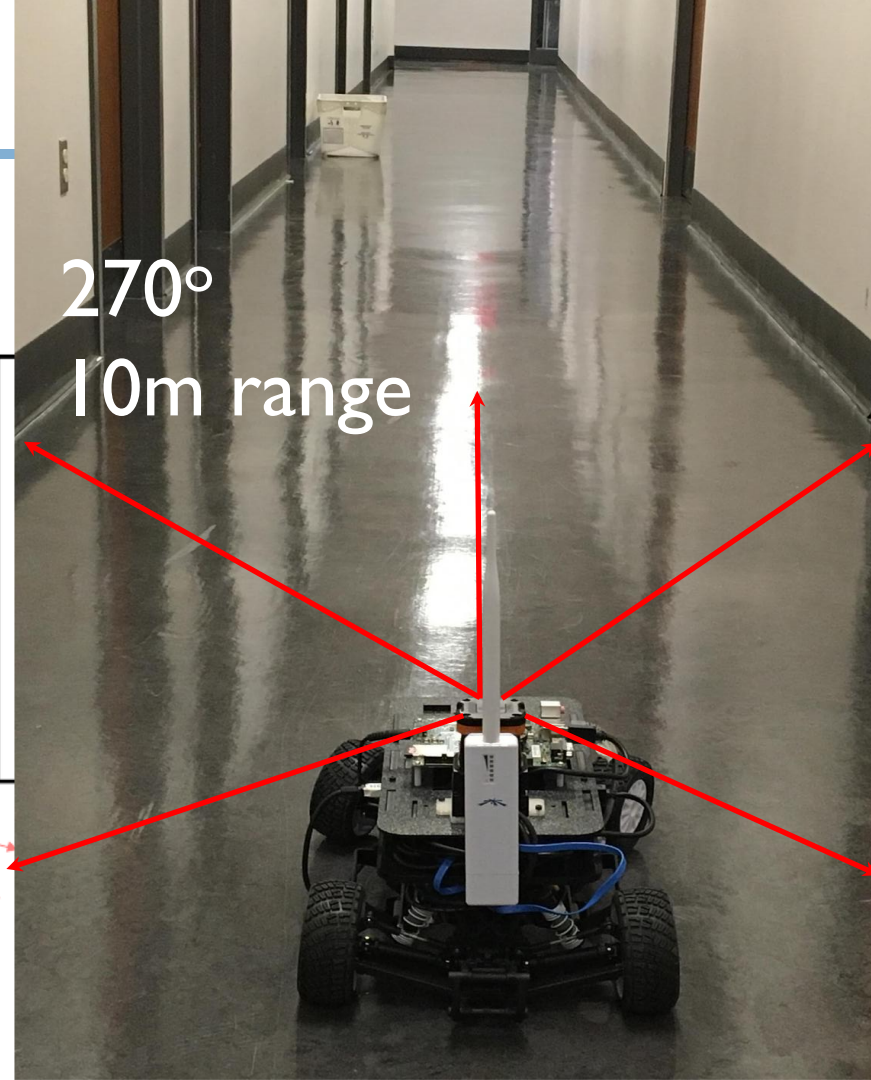
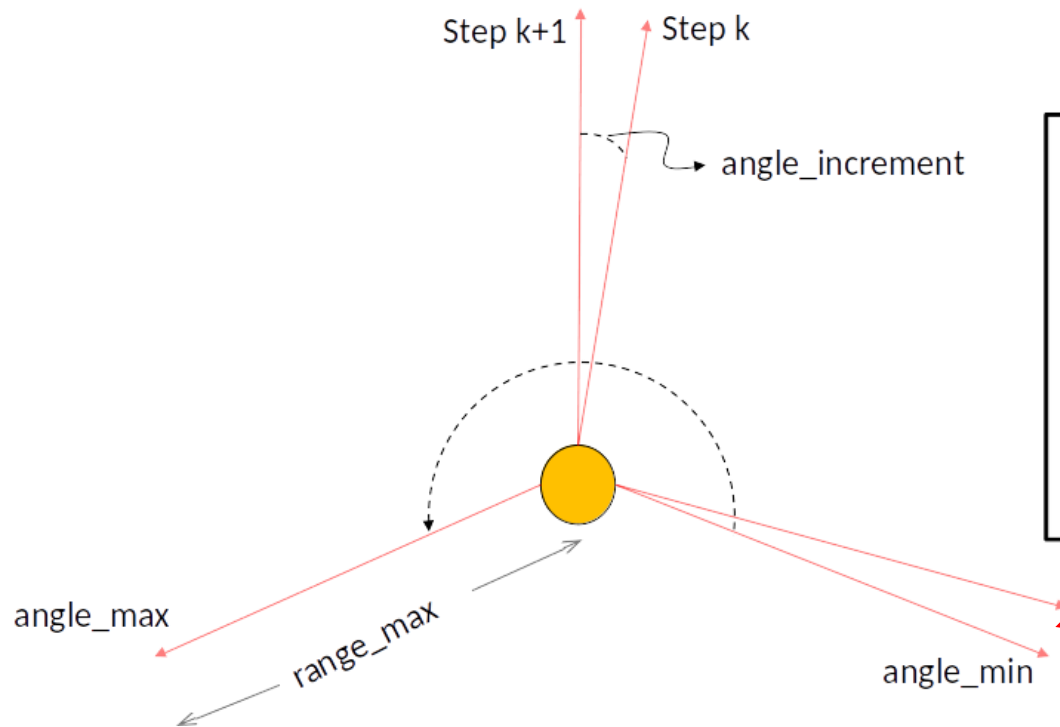
Planar LIDAR

Distance to obstacle = (speed of light * time traveled)/2

- **Maximum range** is constrained by **laser energy**
 - Need a large Signal-to-Noise Ratio (SNR)
- Energy of reflected ray depends upon:
 - Divergence of laser
 - Bouncing reflections
 - Humidity
 - Reflectivity of target
 - Detector sensitivity

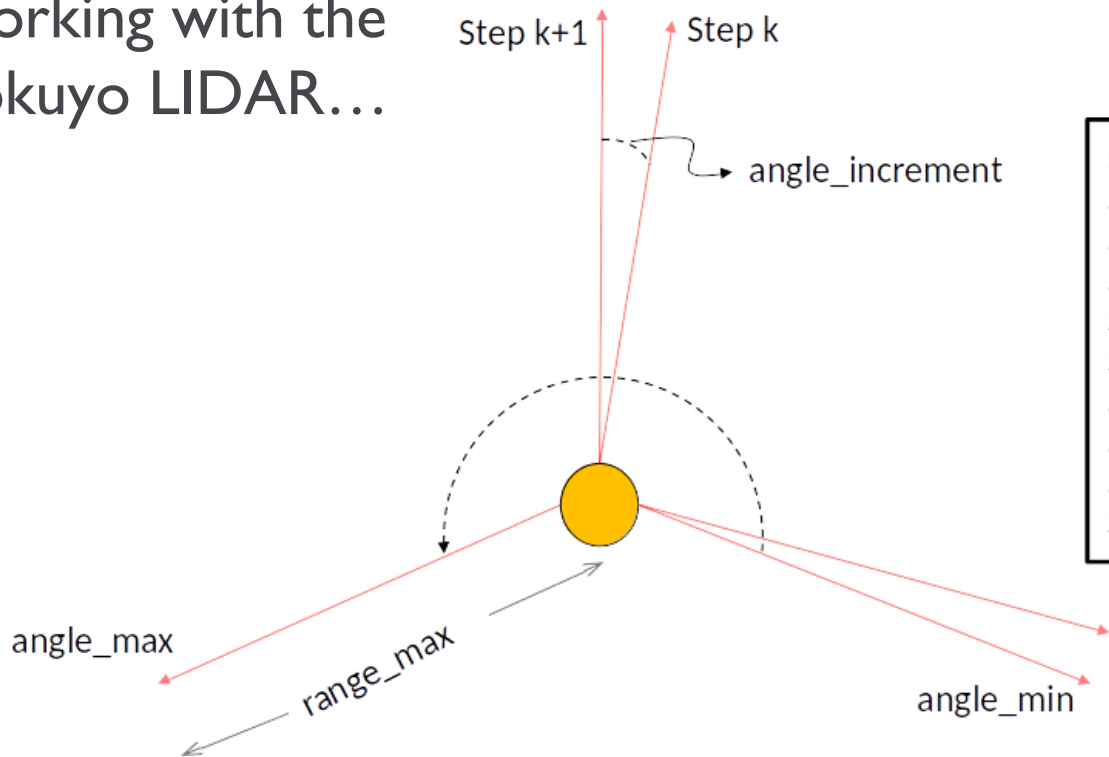


Planar LIDAR



Planar LIDAR

Working with the Hokuyo LIDAR...



```
std_msgs/Header header  
float32 angle_min  
float32 angle_max  
float32 angle_increment  
float32 time_increment  
float32 scan_time  
float32 range_min  
float32 range_max  
float32[] ranges  
float32[] intensities
```

LIDAR: ROS message structure

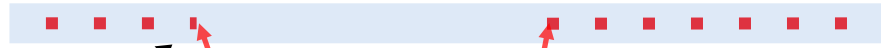
Single scan from a planar laser range-finder

Header header	# timestamp in the header is the acquisition time of # the first ray in the scan.
float32 angle_min	# start angle of the scan [rad]
float32 angle_max	# end angle of the scan [rad]
float32 angle_increment	# angular distance between measurements [rad]
float32 time_increment	# time between measurements [seconds]
float32 scan_time	# time between scans [seconds]
float32 range_min	# minimum range value [m]
float32 range_max	# maximum range value [m]
float32[] ranges	# range data [m] (Note: values < range_min or > range_max should be discarded)
float32[] intensities	# intensity data [device-specific units].

A simple scan sequence

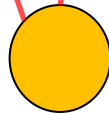
TOP VIEW

Whiteboard

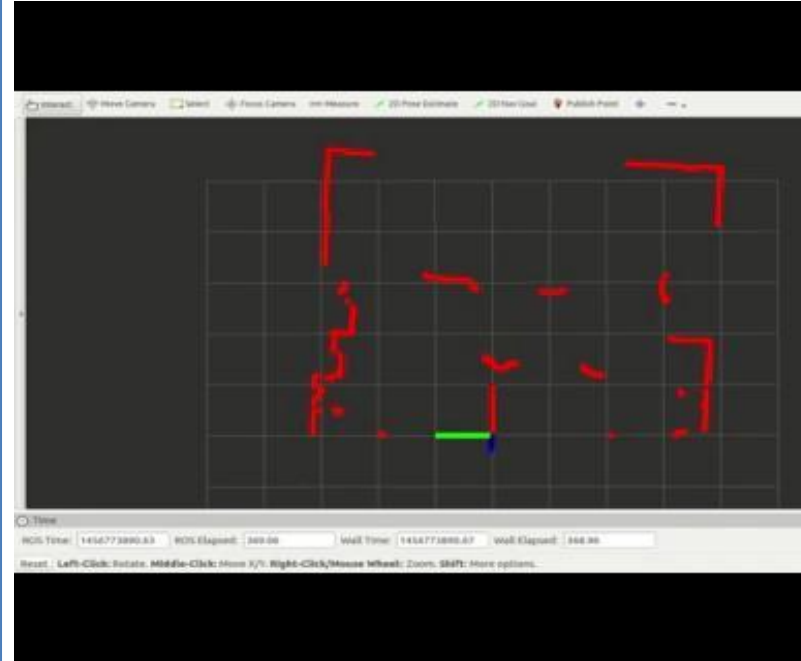


Moving
person

What LIDAR
"sees"



LIDAR



LIDAR: range data

Array $A[1080]$: $A[i]$ is distance measurement of the i -th step.

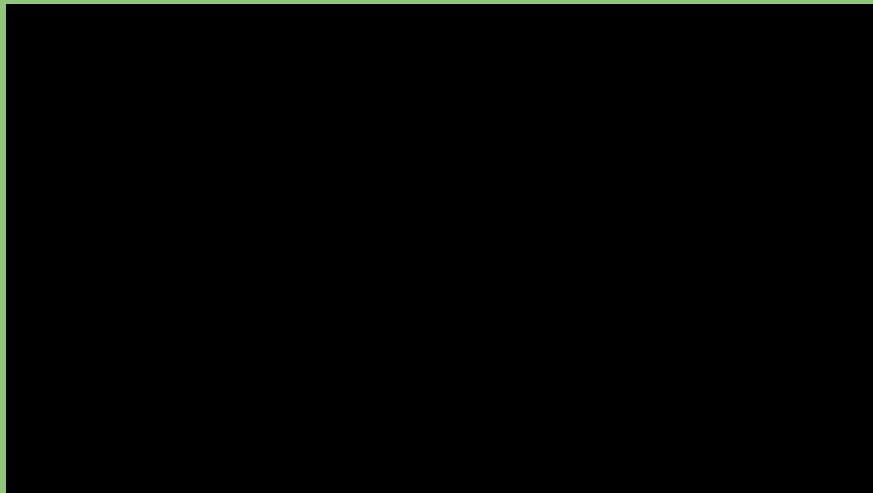
Measurements beyond the min and max range (like inf) *should be discarded*.

[illegible]

Measuring Safety

Safety as an Indicator Function

Safe



Safe AEB with moving vehicles + successful evasion maneuver

Unsafe



No AEB triggered for stationary object in the driving path

From Binary Safety to Continuous Safety...

Safe

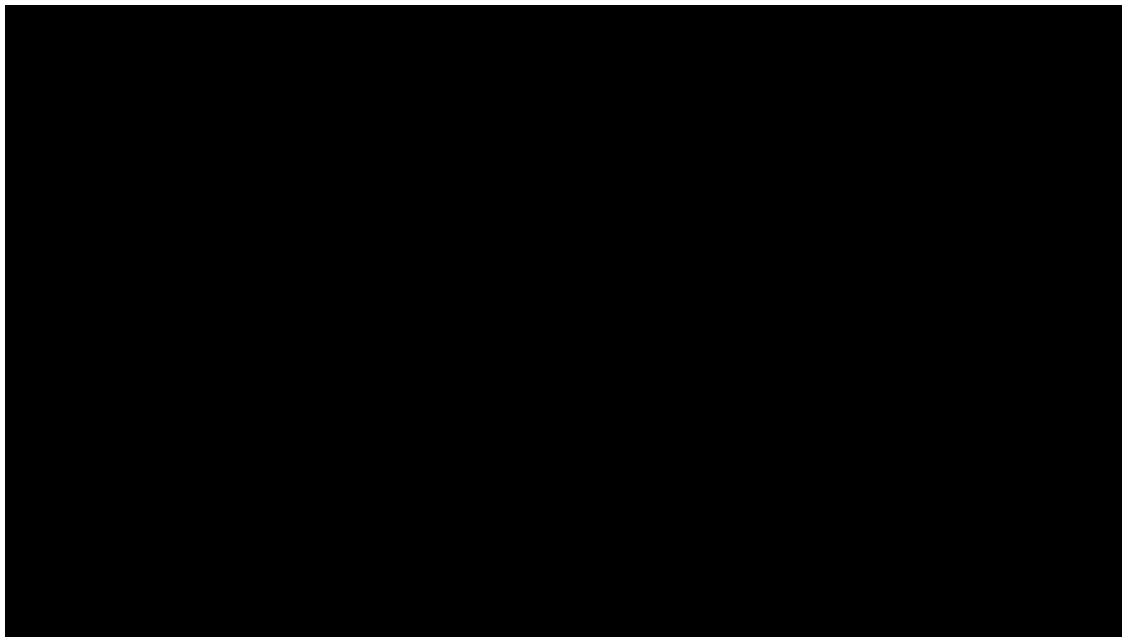
Marginal

Unsafe



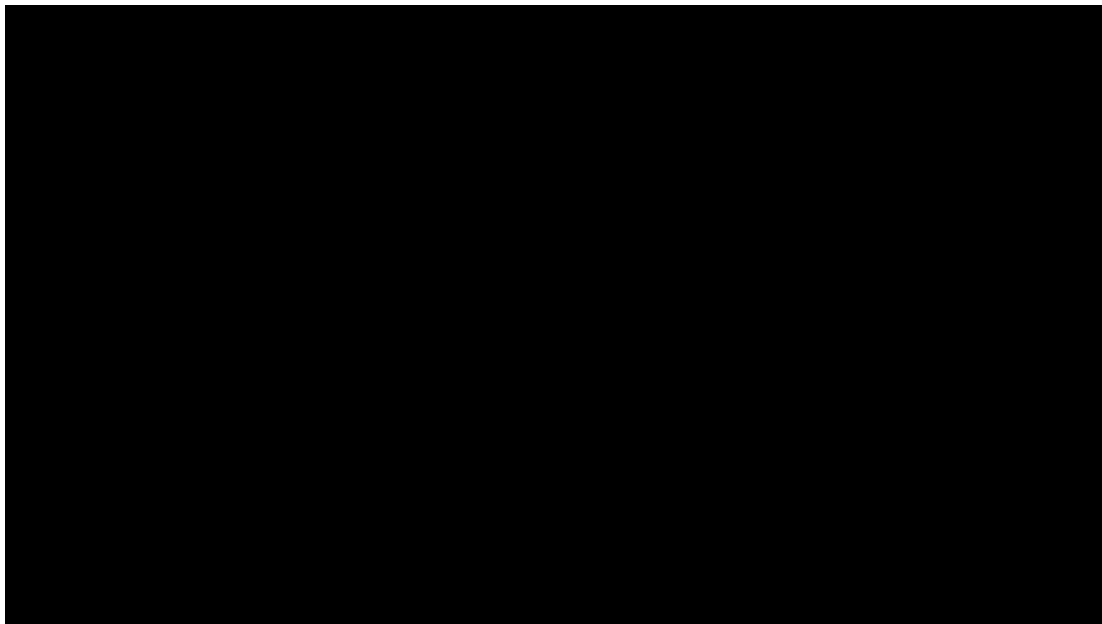
Why not Euclidean Distance?

Pros: Easy to compute, have sensors which can measure accurately with well understood uncertainty.



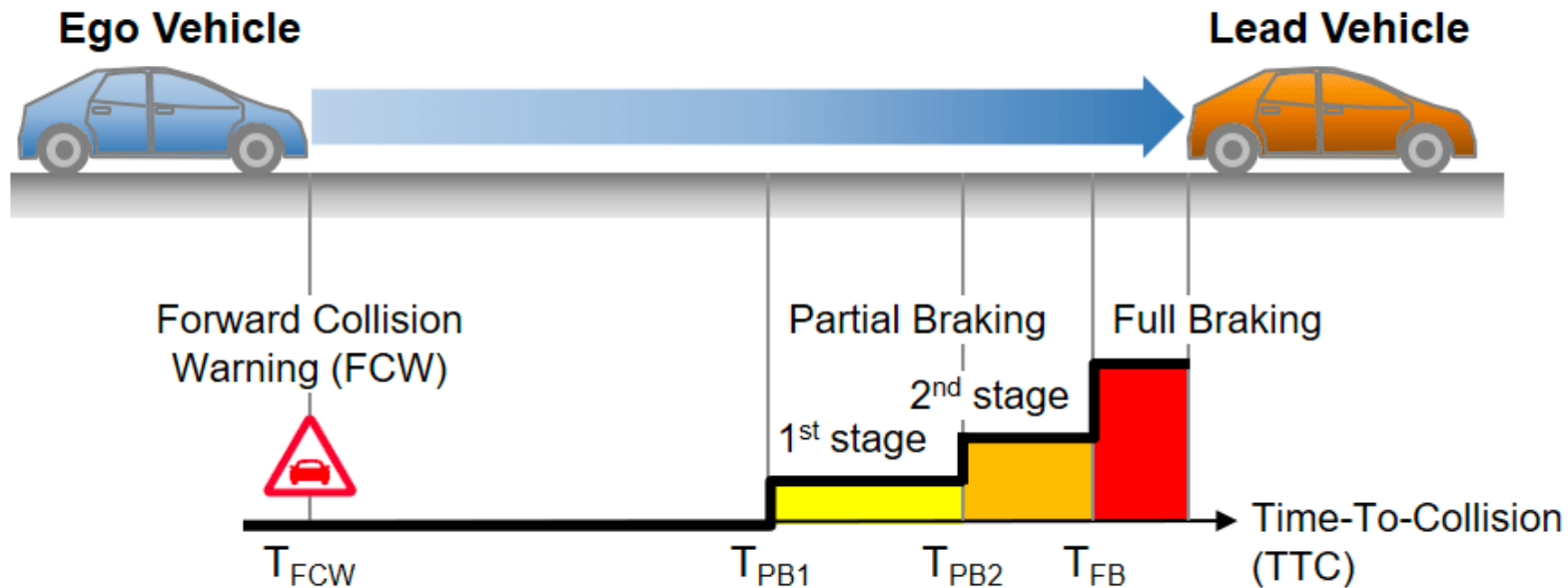
Why not Euclidean Distance?

Cons: Classifies safe things as 'near' unsafe. Slam on the brakes when you are already stopped at T-junction?



Time-to-collision

TTC is defined as the time it would take for the ego-vehicle and an object to intercept one another given that they each maintain their current heading and velocity.



Time-to-collision

TTC is defined as the time it would take for the ego-vehicle and an object to intercept one another given that they each maintain their current heading and velocity.

Note: $TTC_i(t) = \infty$
for $\dot{r}_i(t) \geq 0$

Range between vehicle and object

$$TTC_i(t) = \frac{r_i(t)}{[-\dot{r}_i(t)]_+}$$

Time derivative of range between vehicle and object “range-rate”

Operator defined as:
 $[x]_+ := \max(x, 0)$

How to Compute Range-rate

Range-rate is the time derivative of this distance which is simply computed by projecting the relative velocity of vehicle onto the vector between the vehicles and object's poses...

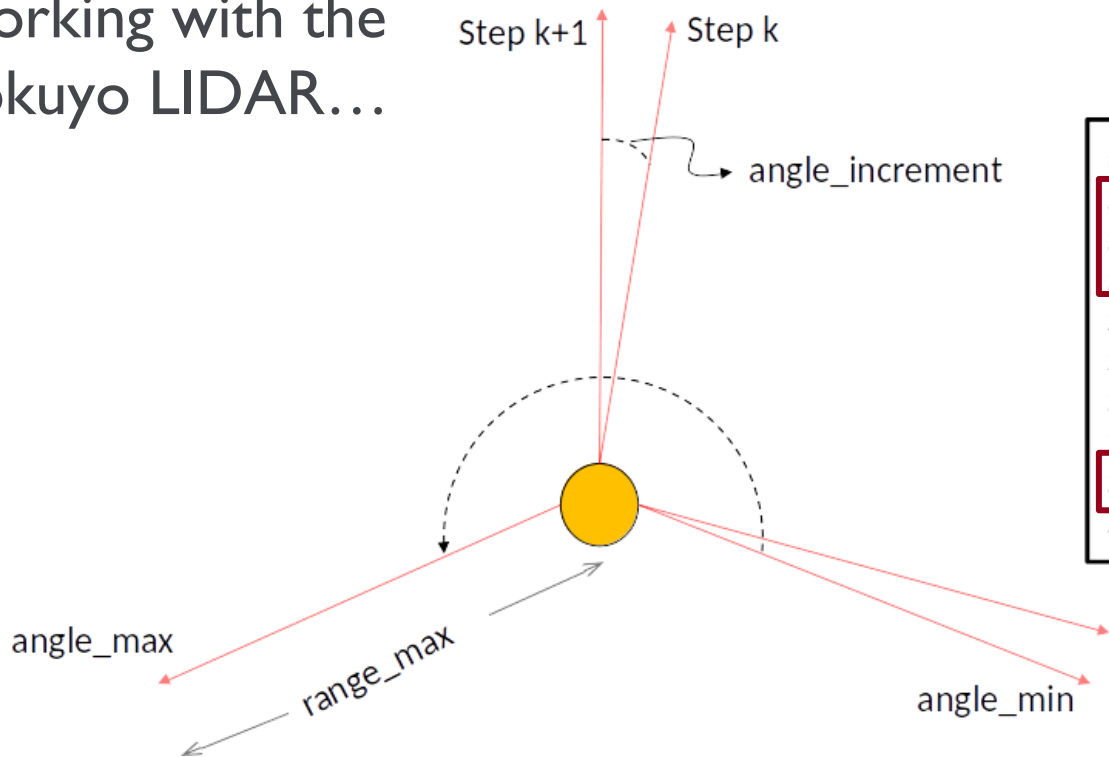
$$\dot{r}_i = v_x \cos(\theta_i)$$

Forward speed in the vehicles reference frame, this is simplified because we are not dealing with moving obstacles

Multiply by the cosine of the beams angle.

Planar LIDAR

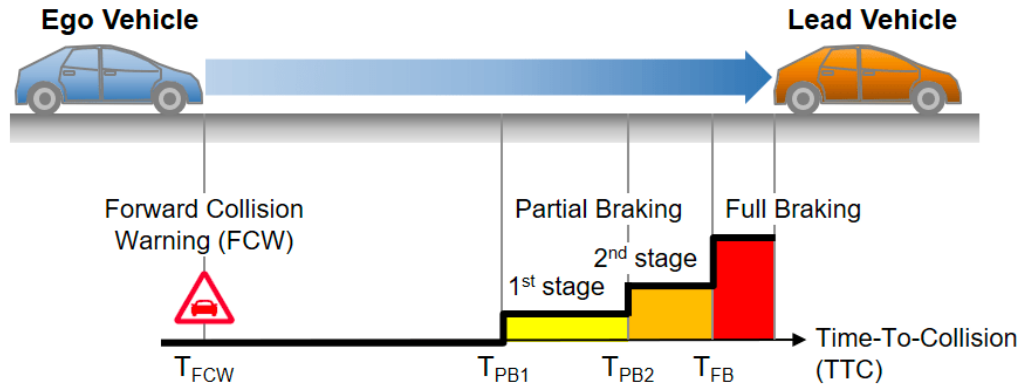
Working with the
Hokuyo LIDAR...



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float32 time_increment  
float32 scan_time  
float32 range_min  
float32 range_max  
float32[] ranges  
float32[] intensities
```

How to Compute Range-rate

If the measured **TTC** falls below the threshold, the system triggers alerts, pre-charges brakes, or initiates autonomous emergency braking



$$T = \min_i TTC_i$$

Questions?